

Unit 1:

Q.1 Define the following terms

- i) Events , ii) Impossible event ,iii) Discrete Random Variable,iv) Elementary event, v) Sample Space
- vi) sure event/ certain event vii) mutually exclusive event viii) mutually exhaustive event
- ix) complement of an event

Q.2 Important theorem on probability.

i) $p(A') = 1 - P(A)$

ii) Addition theorem of probability

Q. 3 $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cap B) = 0.3$. Compute $P(A')$, $P(A \cup B)$, $P(A' \cap B)$, $P(A' \cap B')$ and $P(A' \cup B')$

Q.4 An unbiased coin is tossed three times, what is the probability that

- i) Exactly 2 heads occurred
- ii) At least 2 heads occurred.
- iii) At most 1 head occurred.

Q.5 If two dice are thrown simultaneously, then find the probability that the sum of numbers appeared on the dice is 6 or 7?

Unit 2:

Q.1 Explain the concept of independent of two event. Give an illustration.

Q.2 State and prove Bayes theorem.

Q.3 Explain the concept of conditional probability.

Q.4 A and B are independent with $P(A) = 0.5$, $P(B) = 0.4$. Find (1) $P(A \cup B)$, (ii) $P(A \cap B')$ (iii) $P(A' \cap B)$, (iv) $P(A' \cap B')$.

Q.5 consider a family of two children and assume that all the four possibilities of sex distribution are equally likely

Let A: first child is male child and B: second child is male child. Are A and B independent?

Unit 3:

Q.1 define Discrete random variable, Probability mass function(PDF), Cumulative distribution function (CDF)

Q.2 Properties of distribution function.

Q.3 define mathematical expectation and variance.

Q.4 The mean and variance of marks in Statistics (X) are 60 and 25 respectively. Find the mean and variance of

$$(i) Y = \frac{(X-60)}{5} \quad (ii) Z = \frac{(X-30)}{10}$$

Q.5 Let X be a r.v. with following as the p.m.f.

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Find $E(X)$ and $Var(X)$

Q.6 Let X be a discrete r.v. with mean μ and variance σ^2 . Then, prove

i) $Var(X+b) = Var(X) = \sigma^2$

ii) $Var(aX) = a^2 Var(X) = a^2 \sigma^2$

iv) $Var(aX+b) = a^2 \sigma^2$

Q.7 Define i) Raw moment ii) central moment

Q.8 Define moment generating function and cumulative generating function and their properties.

Unit 4:

Q.1 Theorems on expectation

Q.2 Define Covariance and Correlation coefficient

Q.3 Variance of linear combination . Suppose X and Y two discrete r.v.s Prove that

i) $Var(aX+bY) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov(X,Y)$

ii) $Var(aX-bY) = a^2 Var(X) + b^2 Var(Y) - 2ab Cov(X,Y)$

Q.4 For two discrete r.v.s X,Y; $Var(X)=Var(Y)=1, Cov(X,Y)=1/2$

Find i) $Var(4X-3Y)$

ii) $Var(5X+2Y)$

Q.5 For the following probability distribution ,

	Y	2	3	4
X				
1		0.06	0.15	0.09
2		0.14	0.35	0.21

Find $E(X|Y=2)$ & $V(X|Y=2)$

Q.6 if X and Y are independent discrete r.v such that $Var(X)=4$ and $Var(Y)=3$ compute

i) $Var(2X+5)$

ii) $Var(5X-2Y-1)$

Q.7 For the following probability distribution of (X,Y) compute $\rho(X,Y)$, the correlation coefficient between X and Y .

$X \backslash Y$	0	1	2
0	$1/4$	0	$1/4$
1	$1/8$	$1/8$	$1/4$

Q.6 A r.v. (X,Y) has joint p.m.f. as follows.

$X \backslash Y$	0	1	2
-1	0.1	0.2	0.3
1	0.1	0.1	0.2

i) Marginal distribution of X and Y .

ii) Conditional probability distribution of X given $Y=1$